2. Microscopy

Using the microscope

The setting up of a microscope is a basic skill of microbiology yet it is rarely mastered. Only when it is done properly can the smaller end of the diversity of life be fully appreciated and its many uses in practical microbiology, from aiding identification to checking for contamination, be successfully accomplished. The amount of magnification of which a microscope is capable is an important feature but it is the resolving power that determines the amount of detail that can be seen.

[Appendix 3: Safety resources]

Hints

- ▷ Adjust the iris diaphragm to achieve optimum balance between definition and glare. Do not control light intensity by moving the sub-stage condenser, the position of which should be to focus the light on the specimen. Re-adjust the iris diaphragm for each objective lens.
- ▷ For looking at wet mounts of living specimens of protozoa, algae, moulds and even yeasts, the low power objective lens (×10) is often adequate but also necessary for locating and centering on an area of interest before turning to the high power objective lens (×40). Without altering the focus, turn to the high power lens and then finely re-focus.
- ▷ Use the oil immersion objective lens for examining stained preparations of bacteria. Put one drop of immersion oil onto the preparation; a coverslip is not required. Remove the slide and wipe the oil immersion lens clean at the end of the practical session.

Bacteria and yeast

Yeast can be seen in unstained wet mounts at magnifications ×100. Bacteria are much smaller and can be seen unstained at ×400 but only if the microscope is properly set up and all that is of interest is whether or not they are motile. A magnification of ×1,000 and the use of an oil immersion objective lens for observing stained preparations are necessary for seeing their characteristic shapes and arrangements. The information gained, along with descriptions of colonies, is the starting point for identification of genera and species, but further work involving physiology, biochemistry and molecular biology is then needed.

Moulds

Routine identification of moulds is based entirely on the appearance of colonies to the naked eye and of the mycelium and spores in microscopical preparations.

Mould mycelium and spores can be observed in unstained wet mounts at magnifications of ×100 although direct observations of 'mouldy' material through the lid of a Petri dish or specimen jar at lower magnifications with the plate microscope are also informative (but keep the lid on!). Routine identification of moulds is based entirely on the appearance of colonies to the naked eye and of the mycelium and spores in microscopical preparations.

Hint

Preparing a quick temporary mount for looking at living protozoa or algae.

Take a microscope slide and place 1 or 2 loopfuls of the sample in the centre. Place a coverslip over the sample, avoiding air bubbles. Seal each edge of the coverslip in turn with a thin film of Vaseline from the warmed end of a microscope slide. Keep microscope lamps as dim as possible to stop the culture from drying out.

Protozoa and algae

Protozoa and algae are large organisms and therefore are readily visible at a magnification of ×10 to ×100 in unstained wet mounts. A magnification of ×100 is advantageous for observing natural samples that contain a variety of organisms, particularly as many are very motile. Identification of algae and protozoa is based entirely on their microscopical appearance. The common algae are green and non-motile; diatoms have a brown, sculptured outer layer of silica and move slowly. Protozoa are colourless and most are motile. Hay infusions and cloudy vase water are rich in algae and protozoa, but clear samples of water are rarely rewarding.

For further information contact Sciento and see *Appendix 4: Suppliers of cultures and equipment.*